

FORMULAE

Appendix - A

◆ Derivatives of standard functions

$$1. \quad \frac{d}{dx}(c) = 0$$

$$2. \quad (a) \quad \frac{d}{dx}(x^n) = n x^{n-1} \qquad (b) \quad \frac{d}{dx}(ax+b)^n = n(ax+b)^{n-1}(a)$$

$$3. \quad (a) \quad \frac{d}{dx}(e^x) = e^x \quad (b) \quad \frac{d}{dx}(e^{ax}) = (e^{ax})(a) \quad (c) \quad \frac{d}{dx}(e^{ax+b}) = (e^{ax+b})(a)$$

$$4. \quad \frac{d}{dx}(a^x) = a^x \log a$$

$$5. \quad (a) \quad \frac{d}{dx}(\sin x) = \cos x \qquad (b) \quad \frac{d}{dx}(\sin ax) = (\cos ax)(a)$$

$$(c) \quad \frac{d}{dx}[\sin(ax+b)] = [\cos(ax+b)](a)$$

$$6. \quad (a) \quad \frac{d}{dx}(\cos x) = -\sin x \qquad (b) \quad \frac{d}{dx}(\cos ax) = (-\sin ax)(a)$$

$$(c) \quad \frac{d}{dx}[\cos(ax+b)] = [-\sin(ax+b)](a)$$

$$7. \quad (a) \quad \frac{d}{dx}(\tan x) = \sec^2 x \qquad (b) \quad \frac{d}{dx}(\tan ax) = (\sec^2 ax)(a)$$

$$(c) \quad \frac{d}{dx}[\tan(ax+b)] = [\sec^2(ax+b)](a)$$

$$8. \quad (a) \quad \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x \qquad (b) \quad \frac{d}{dx}(\cot ax) = [-\operatorname{cosec}^2 ax](a)$$

$$(c) \quad \frac{d}{dx}[\cot(ax+b)] = [-\operatorname{cosec}^2(ax+b)](a)$$

$$9. \quad (a) \quad \frac{d}{dx}(\sec x) = \sec x \tan x \qquad (b) \quad \frac{d}{dx}[\sec ax] = [\sec ax \tan ax](a)$$

$$(c) \quad \frac{d}{dx}[\sec(ax+b)] = [\sec(ax+b) \tan(ax+b)](a)$$

$$10. \quad (a) \quad \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x \quad (b) \quad \frac{d}{dx}(\operatorname{cosec} ax) = [-\operatorname{cosec} ax \cot ax](a)$$

$$(c) \quad \frac{d}{dx}[\operatorname{cosec}(ax+b)] = [-\operatorname{cosec}(ax+b) \cot(ax+b)](a)$$

$$11. \quad \frac{d}{dx}(\log x) = \frac{1}{x}, \quad x \neq 0$$

$$12. \quad \frac{d}{dx}(\log_a x) = \frac{\log_a e}{x}, \quad x \neq 0$$

$$13. \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$14. \quad \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

- $$15. \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$
- $$16. \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$
- $$17. \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$
- $$18. \frac{d}{dx} \operatorname{cosec}^{-1} x = -\frac{1}{x\sqrt{x^2-1}}$$
- $$19. \frac{d}{dx} \sinh x = \cosh x$$
- $$20. \frac{d}{dx} \cosh x = \sinh x$$
- $$21. \frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$
- $$22. \frac{d}{dx} \coth x = -\operatorname{cosech}^2 x$$
- $$23. \frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$
- $$24. \frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \coth x$$
- $$25. \frac{d}{dx} \sinh^{-1} x = \frac{1}{\sqrt{1+x^2}}$$
- $$26. \frac{d}{dx} \cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}$$
- $$27. \frac{d}{dx} \tanh^{-1} x = \frac{1}{1-x^2}$$
- $$28. \frac{d}{dx} \coth^{-1} x = -\frac{1}{x^2-1}$$
- $$29. \frac{d}{dx} \operatorname{sech}^{-1} x = -\frac{1}{x\sqrt{1-x^2}}$$
- $$30. \frac{d}{dx} \operatorname{cosech}^{-1} x = -\frac{1}{x\sqrt{x^2+1}}$$

Appendix - B

◆ Integrals of standard functions

- $$1. \int x^n dx = \frac{x^{n+1}}{n+1}, n \neq -1$$
- $$2. \int \frac{dx}{x} = \log x, x \neq 0$$
- $$3. \int e^x dx = e^x$$
- $$4. \int a^x dx = \frac{a^x}{\log a}, a \neq 1$$
- $$5. (a) \int \sin x dx = -\cos x$$
- $$(b) \int \sin ax dx = -\frac{1}{a} \cos ax$$
- $$(c) \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b)$$
- $$6. (a) \int \cos x dx = \sin x$$
- $$(b) \int \cos ax dx = \frac{1}{a} \sin ax$$
- $$(c) \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b)$$
- $$7. (a) \int \tan x dx = \log [\sec x]$$
- $$(b) \int \tan ax dx = \frac{1}{a} \log [\sec ax]$$
- $$(c) \int \tan(ax+b) dx = \frac{1}{a} \log [\sec(ax+b)]$$

$$8. \quad (a) \int \cot x \, dx = \log \sin x \quad (b) \int \cot ax \, dx = \frac{1}{a} \log \sin ax$$

$$(c) \int \cot (ax + b) \, dx = \frac{1}{a} \log [\sin (ax + b)]$$

$$9. \quad (a) \int \sec x \, dx = \log [\sec x + \tan x]$$

$$(b) \int \sec ax \, dx = \frac{1}{a} \log [\sec ax + \tan ax]$$

$$(c) \int \sec (ax + b) \, dx = \frac{1}{a} \log [\sec (ax + b) + \tan (ax + b)]$$

$$10. \quad (a) \int \operatorname{cosec} x \, dx = \log [\operatorname{cosec} x - \cot x]$$

$$(b) \int \operatorname{cosec} ax \, dx = \frac{1}{a} \log [\operatorname{cosec} ax - \cot ax]$$

$$(c) \int \operatorname{cosec} (ax + b) \, dx = \frac{1}{a} \log [\operatorname{cosec} (ax + b) - \cot (ax + b)]$$

$$11. \int \sec^2 x \, dx = \tan x \quad 12. \int \operatorname{cosec}^2 x \, dx = -\cot x$$

$$13. \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x \quad 14. \int \sec x \tan x \, dx = \sec x$$

$$15. \int \sinh x \, dx = \cosh x \quad 16. \int \cosh x \, dx = \sinh x$$

$$17. \int \tanh x \, dx = \log \cosh x \quad 18. \int \coth x \, dx = \log \sinh x$$

$$19. \int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x \quad 20. \int \operatorname{sech}^2 x \, dx = \tanh x$$

$$21. \int \operatorname{cosech} x \coth x \, dx = -\operatorname{cosech} x \quad 22. \int \operatorname{cosech}^2 x \, dx = -\coth x$$

$$23. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$24. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left(\frac{x-a}{x+a} \right), (x > a), \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left(\frac{a+x}{a-x} \right), (x < a)$$

$$25. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$$

$$26. \int \sqrt{a^2 - x^2} \, dx = \frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$27. \int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1} \frac{x}{a} = \log \frac{x + \sqrt{x^2 + a^2}}{a}$$

$$28. \int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} = \log \frac{x + \sqrt{x^2 - a^2}}{a}$$

$$29. \int \sqrt{a^2 + x^2} dx = \frac{x \sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \log \frac{x + \sqrt{x^2 + a^2}}{a}$$

$$= \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \log (x + \sqrt{x^2 + a^2})$$

$$= \frac{x \sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \sinh^{-1} \left(\frac{x}{a} \right)$$

$$30. \int \sqrt{x^2 - a^2} dx = \frac{x \sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \log \frac{x + \sqrt{x^2 - a^2}}{a}$$

$$= \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log (x + \sqrt{x^2 - a^2})$$

$$= \frac{x \sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \cosh^{-1} \left(\frac{x}{a} \right)$$

$$31. \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x$$

$$32. \int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1} x$$

$$33. \int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$34. \int \frac{-1}{1+x^2} dx = \cot^{-1} x$$

$$35. \int \frac{1}{x \sqrt{x^2 - 1}} dx = \sec^{-1} x$$

$$36. \int \frac{-1}{x \sqrt{x^2 - 1}} dx = \operatorname{cosec}^{-1} x$$

$$37. \int \frac{1}{\sqrt{1+x^2}} dx = \sinh^{-1} x$$

$$38. \int \frac{1}{\sqrt{x^2 - 1}} dx = \cosh^{-1} x$$

$$39. \int \frac{1}{1-x^2} dx = \tanh^{-1} x$$

$$40. \int \frac{-1}{x^2 - 1} dx = \coth^{-1} x$$

$$41. \int \frac{-1}{x \sqrt{1-x^2}} dx = \operatorname{sech}^{-1} x$$

$$42. \int \frac{-1}{x \sqrt{x^2 + 1}} dx = \operatorname{cosech}^{-1} x$$

$$43. \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$44. \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$45. \int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2}$$

$$46. \int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

$$47. \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x}$$

$$48. \int \log x dx = x \log x - x$$

$$49. I_n = \int_0^{\frac{\pi}{2}} \cos^n \theta d\theta = \int_0^{\frac{\pi}{2}} \sin^n \theta d\theta$$

$$\text{If } n \text{ is odd } I_n = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3} \cdot 1$$

$$\text{If } n \text{ is even } I_n = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$$

$$50. \int_0^{\pi/2} \sin^m x \cos^n x dx$$

$$= \frac{m-1}{m+n} \frac{m-3}{m+n-2} \cdots \frac{2}{3+n} \frac{1}{1+n}$$

[if m is odd, n may be even or odd]

$$= \frac{m-1}{m+n} \frac{m-3}{m+n-2} \cdots \frac{1}{2+n} \frac{n-1}{n} \frac{n-3}{n-2} \cdots \frac{2}{3} \cdot 1$$

[if m is even, n is odd]

$$= \frac{m-1}{m+n} \frac{m-3}{m+n-2} \cdots \frac{1}{2+n} \frac{n-1}{n} \frac{n-3}{n-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$$

[if m is even, n is even]

$$51. \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is an even function}$$

$$= 0 \quad \text{if } f(x) \text{ is an odd function}$$

$$52. \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(2a-x) = f(x) \text{ [even like function]}$$

$$= 0 \quad \text{if } f(2a-x) = -f(x) \text{ [odd like function]}$$

$$53. \int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx, \quad a < c < b$$

$$54. \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$55. \int u dv = uv - \int v du$$

$$56. \int uv dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots \quad [\text{Bernoulli}]$$

Appendix - C

◆ Trigonometry

I. Inter-relations :

1. $\sin \theta = \frac{1}{\operatorname{cosec} \theta}$

2. $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

3. $\cos \theta = \frac{1}{\sec \theta}$

4. $\sec \theta = \frac{1}{\cos \theta}$

5. $\tan \theta = \frac{1}{\cot \theta}$

6. $\cot \theta = \frac{1}{\tan \theta}$

7. $\frac{\sin \theta}{\cos \theta} = \tan \theta$

8. $\frac{\cos \theta}{\sin \theta} = \cot \theta$

II. Identities

1. $\sin^2 \theta + \cos^2 \theta = 1$

2. $1 + \tan^2 \theta = \sec^2 \theta$

3. $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

III. Radian measure

$$\pi \text{ radians} = 180^\circ$$

IV. Trigonometric ratios for certain standard angles

θ	0°	$30^\circ (\pi/6)$	$45^\circ (\pi/4)$	$60^\circ (\pi/3)$	$90^\circ (\pi/2)$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞ (infinity)

$\cot \theta$, $\sec \theta$, $\operatorname{cosec} \theta$ are respectively the reciprocals of $\tan \theta$, $\cos \theta$, $\sin \theta$

V. Allied angles

Trigonometrical ratios of $90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$, $360^\circ \pm \theta$ in terms of those of θ can be found easily by the following rule known as A - S - T - C rule.

- (i) When the angle is $90^\circ \pm \theta$ or $270^\circ \pm \theta$ the trigonometrical ratio changes from sine to cosine and vice versa. Also $\tan \rightleftharpoons \cot$, $\sec \rightleftharpoons \operatorname{cosec}$
- (ii) When the angle is $180^\circ \pm \theta$ or $360^\circ \pm \theta$ the trigonometrical ratio remains the same. i.e., $\sin \rightarrow \sin$, $\cos \rightarrow \cos$, etc.

(iii) In each case the sign + or - is pre multiplied by the A-S-T-C quadrant rule :

S	A	A : All ratios are +ve in the I quadrant
II ($90^\circ - 180^\circ$)	I ($0^\circ - 90^\circ$)	S : sin is +ve in the II quadrant
T	C	T : tan is +ve in the III quadrant.
III ($180^\circ - 270^\circ$)	IV ($270^\circ - 360^\circ$)	C : cos is +ve in the IV quadrant

Note : $\sin(-\theta) = -\sin \theta$, $\cos(-\theta) = \cos \theta$,
 $\sin(n \cdot 2\pi + \theta) = \sin \theta$, $\cos(n \cdot 2\pi + \theta) = \cos \theta$

Example 1 : $\sin(90^\circ - \theta) = \cos \theta$, $\cos(90^\circ + \theta) = -\sin \theta$

$\sin(180^\circ - \theta) = \sin \theta$, $\tan(180^\circ + \theta) = \tan \theta$ etc.

Example 2 : $\sin(135^\circ) = \sin(90^\circ + 45^\circ) = \cos 45^\circ = \frac{1}{\sqrt{2}}$

$\tan(315^\circ) = \tan(270^\circ + 45^\circ) = -\cot 45^\circ = -1$

$\cos(225^\circ) = \cos(180^\circ + 45^\circ) = -\cos 45^\circ = -\frac{1}{\sqrt{2}}$

$\sin(750^\circ) = \sin(2 \times 360^\circ + 30^\circ) = \sin 30^\circ = \frac{1}{2}$ etc.

VI. Compound angle formulae

(i) $\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\sin(A - B) = \sin A \cos B - \cos A \sin B$

(ii) $\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

(iii) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

VII. Formulae to convert a product into sum or difference

(iv) $\left. \begin{array}{l} \sin A \cos B \\ \text{or } \cos B \sin A \end{array} \right\} = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$

$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$, $\sin(-\theta) = -\sin \theta$

$$(v) \cos A \cos B = \frac{1}{2} [\cos (A+B) + \cos (A-B)]$$

$$(vi) \sin A \sin B = \frac{1}{2} [\cos (A-B) - \cos (A+B)], \quad \cos (-\theta) = \cos \theta$$

VIII. Particular cases of formula (i) to (vi)

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 1 - 2 \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$1 + \cos 2A = 2 \cos^2 A$$

$$1 - \cos 2A = 2 \sin^2 A$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\sin A = 2 \sin (A/2) \cos (A/2)$$

$$\cos A = \cos^2 (A/2) - \sin^2 (A/2)$$

$$= 1 - 2 \sin^2 (A/2)$$

$$= 2 \cos^2 (A/2) - 1$$

$$1 + \cos A = 2 \cos^2 \frac{A}{2}$$

$$1 - \cos A = 2 \sin^2 \frac{A}{2}$$

$$\sin A = \frac{2 \tan (A/2)}{1 + \tan^2 (A/2)}$$

$$\cos A = \frac{1 - \tan^2 (A/2)}{1 + \tan^2 (A/2)}$$

IX. Formulae to convert a sum or difference into product.

$$(vii) \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$(viii) \sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$(ix) \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$(x) \cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

X. Hyperbolic functions

We have already said that 'e' whose value is approximately 2.7 is called the exponential constant. Further if $\log_e y = x$ then $y = e^x$ is called as the exponential function. Hyperbolic functions are defined in terms of exponential function as below.

$$\sinh x = \frac{e^x - e^{-x}}{2}; \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\text{Also, } \tanh x = \frac{\sinh x}{\cosh x}; \coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}; \operatorname{cosech} x = \frac{1}{\sinh x}$$

XI. Important hyperbolic identities

$$\begin{aligned} \text{(i) } \cosh^2 x - \sinh^2 x &= 1 & \text{(ii) } 1 - \tanh^2 x &= \operatorname{sech}^2 x \\ \text{(iii) } \coth^2 x - 1 &= \operatorname{cosech}^2 x & \text{(iv) } \cosh^2 x + \sinh^2 x &= \cosh 2x \\ \text{(v) } 2 \sinh x \cosh x &= \sinh 2x \end{aligned}$$

XII. Relationship between trigonometric and hyperbolic functions

$$\text{We have } \sin x = \frac{e^{ix} - e^{-ix}}{2i} \text{ and } \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\text{Now } \sin(ix) = \frac{e^{-x} - e^x}{2i} = (-1) \frac{(e^x - e^{-x})}{2i} = \frac{i^2 (e^x - e^{-x})}{2i}$$

$$\therefore \sin(ix) = i \cdot \frac{e^x - e^{-x}}{2} \quad (\text{i.e.,}) \sin(ix) = i \sinh x$$

$$\text{Also } \cos(ix) = \frac{e^{-x} + e^x}{2} = \cosh x \quad (\text{i.e.,}) \cos(ix) = \cosh x$$

XIII. The following are some of the established standard limits.

$$\text{(i) } \lim_{x \rightarrow a} \left(\frac{x^n - a^n}{x - a} \right) = na^{n-1}, \quad n \text{ is any rational number.}$$

$$\text{(ii) } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{(iii) } \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\text{(iv) } \lim_{x \rightarrow 0} (1+x)^{1/x} = e \quad \text{or} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$$

$$\text{(v) } \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x$$

$$\text{(vi) } \lim_{x \rightarrow \infty} x^{1/x} = 1$$

Appendix - D

I. Some standard forms of the Binomial Expansion

For all values of n , when $|x| < 1$, we have

$$1. \quad (1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \dots$$

$$2. \quad (1-x)^n = 1 - \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 - \dots$$

$$3. \quad (1-x)^{-n} = 1 + \frac{n}{1!}x + \frac{n(n+1)}{2!}x^2 + \dots$$

$$4. \quad (1+x)^{-n} = 1 - \frac{n}{1!}x + \frac{n(n+1)}{2!}x^2 - \dots$$

$$5. \quad (1-x)^{-1} = 1 + x + x^2 + \dots$$

$$6. \quad (1+x)^{-1} = 1 - x + x^2 - \dots$$

$$7. \quad (1-x)^{-2} = 1 + 2x + 3x^2 + \dots$$

$$8. \quad (1+x)^{-2} = 1 - 2x + 3x^2 - \dots$$

$$9. \quad (1+x)^{-3} = \frac{1}{2} [1.2 - 2.3x + 3.4x^2 - \dots]$$

II. Resolution of Rational into partial fraction.

Case (i) : Let $\frac{P(x)}{Q(x)}$ be a proper fraction

[i.e., degree of $P(x)$ is less than degree of $Q(x)$]

Type 1: Suppose $Q(x)$ is factorizable into non-repeated linear factors

$$a_1x + b_1, a_2x + b_2, a_3x + b_3$$

$$(i.e.,) Q(x) = (a_1x + b_1)(a_2x + b_2)(a_3x + b_3)$$

$$\frac{P(x)}{Q(x)} = \frac{A}{a_1x + b_1} + \frac{B}{a_2x + b_2} + \frac{C}{a_3x + b_3}$$

where A, B, C are constants.

$$\frac{1}{(x-1)(x+1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+2}$$

Type 2 : Suppose $Q(x)$ contain a repeated linear factor of the form

$$(ax + b)^r$$

$$\frac{P(x)}{Q(x)} = \frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \frac{A_3}{(ax + b)^3} + \dots + \frac{A_r}{(ax + b)^r}$$

$$\text{Example : } \frac{2x+3}{(x+5)^3} = \frac{A}{x+5} + \frac{B}{(x+5)^2} + \frac{C}{(x+5)^3}$$

Type 3 : $Q(x)$ contain a non-repeated quadratic function.

$ax^2 + bx + c$ then

$$\frac{P(x)}{Q(x)} = \frac{Ax + B}{ax^2 + bx + c}$$

$$\text{Example : } \frac{x+3}{(x^2+1)(x-3)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-3}$$

Type 4 : $Q(x)$ contain a repeated quadratic factor $(ax^2 + bx + c)^2$

$$\frac{P(x)}{Q(x)} = \frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2}$$

Case (ii) : Let $\frac{P(x)}{Q(x)}$ be an improper fraction

[i.e., degree of $P(x)$ is greater than or equal to degree of $Q(x)$]

In this case we can write

$$\frac{P(x)}{Q(x)} = f(x) + \frac{R(x)}{Q(x)} \text{ where } f(x) \text{ is the quotient.}$$

(Integral function) and $\frac{R(x)}{Q(x)}$ is a proper fraction.

III. Seven Indeterminates are

$$1. \frac{0}{0} \quad 2. \frac{\infty}{\infty} \quad 3. 0 \times \infty \quad 4. \infty - \infty$$

$$5. 1^\infty \quad 6. \infty^0 \quad 7. 0^0$$

$$\text{IV. } 1. \log_a 1 = 0 \quad 2. \log_a a = 1$$

$$3. \frac{a}{0} = \infty \text{ [where } a \text{ is any real number except zero]}$$

$$4. \frac{0}{a} = 0 \quad 5. e = 2.7183 \text{ (approx)}$$

$$6. \pi = 3.14 \text{ (approx)} \quad 7. \log_a 0 = -\infty \text{ (} a > 1 \text{)}$$

$$8. \log(mn) = \log m + \log n$$

$$9. \log\left(\frac{m}{n}\right) = \log m - \log n$$

$$10. \log(m^n) = n \log m$$

S.No.	Curve	Cartesian equation	Parametric equations
1.	Parabola	1. $y^2 = 4ax$ 2. $x^2 = 4ay$	1. $x = at^2$, $y = 2at$ 2. $x = 2at$, $y = at^2$
2.	Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$x = a \cos \theta$, $y = b \sin \theta$
3.	Circle	$x^2 + y^2 = a^2$	$x = a \cos \theta$, $y = a \sin \theta$
4.	Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$x = a \sec \theta$, $y = b \tan \theta$
5.	Rectangular hyperbola	1. $x^2 - y^2 = a^2$ 2. $xy = c^2$	$x = a \sec \theta$, $y = a \tan \theta$ $x = ct$, $y = \frac{c}{t}$
6.	Astroid	$x^{2/3} + y^{2/3} = a^{2/3}$	$x = a \cos^3 \theta$, $y = a \sin^3 \theta$
7.	Cycloid	--	1. $x = a(\theta + \sin \theta)$ $y = a(1 - \cos \theta)$ 2. $x = a(\theta + \sin \theta)$ $y = a(1 + \cos \theta)$ 3. $x = a(\theta - \sin \theta)$ $y = a(1 - \cos \theta)$ 4. $x = a(\theta - \sin \theta)$ $y = a(1 + \cos \theta)$